

WGS/WES best practices

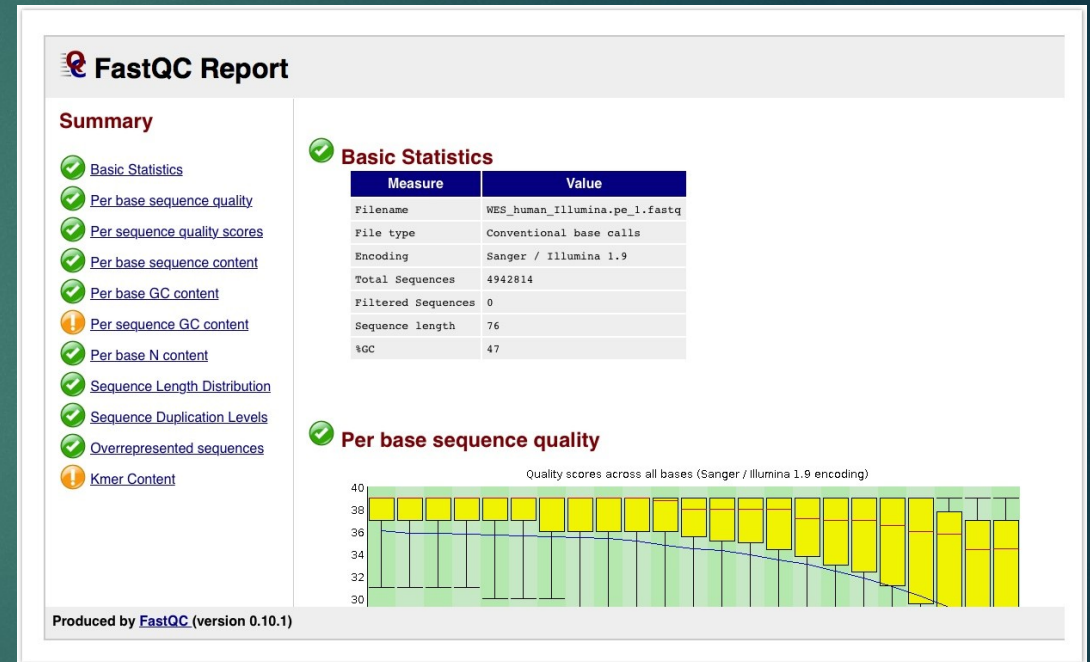
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Experimental design

- ▶ Coverage for germline mutations
 - ▶ WES: 100X coverage, where on average, each base in the exonic region can be read 100 times
 - ▶ WGS: 30X coverage
- ▶ Coverage for somatic (tumor/normal)
 - ▶ WES: 100X-150X
 - ▶ WGS: 100X for tumor samples
- ▶ If looking at somatic mutations, must have tumor and normal DNA from same patient.
- ▶ Sample size depends heavily on study objectives, effect sizes, and statistical power
 - ▶ Common variants require fewer samples.
 - ▶ Strong effect (eg. Mendelian Disease) requires fewer samples

Best practices during quality control

- ▶ **Purpose:** Assess the quality of the reads and ensure that the sequencing data is of high quality and reliable for downstream analysis.
- ▶ **FastQC:** Identifies potential issues such as poor read quality, adapter contamination, and sequence biases.
- ▶ **Fastq_screen:** Aligns reads to multiple reference genomes/databases. Good way to detect contamination.
- ▶ **Adapter trimming:** Adapter trimming is optional, but often recommended. CutAdapt is the most common tool used for adapter trimming.



WES vs. WGS

- ▶ WES focuses on the coding regions of the genome.
 - ▶ Often used in smaller studies where cost is a concern.
 - ▶ Difficult time capturing structural variants, mutations in non-coding region.
 - ▶ Used as first-line, quicker analysis compared to WGS
- ▶ WGS sequences the entire genome
 - ▶ Used when cost and time aren't a problem
 - ▶ Captures structural variants and mutations in non-coding regions

Best practices during alignment

- ▶ BWA (BWA-MEM2) – Most popular for WGS/WES
 - ▶ Pros: Accurate, works well with GATK
 - ▶ Cons: Can be slow and computationally intensive
- ▶ Bowtie2 – High speed and memory efficient aligner.
 - ▶ Pros: Fast, not as computationally intensive
 - ▶ Lower sensitivity than BWA-MEM
- ▶ Novoalign – Closed source, high accuracy
 - ▶ Pros: Very accurate and reliable
 - ▶ Cons: Can be slow and requires a license
- ▶ DRAGEN – Closed source, high accuracy, fast
 - ▶ Pros: Accuracy, speed, alignment and variant/copy number calling all in one step
 - ▶ Cons: Requires a license

Studies comparing aligners

- ▶ [Betschart et al](#) – Benchmark study comparing BWA-MEM with DRAGEN
 - ▶ Study showed that alignment and mappings step was more accurate and faster using DRAGEN vs. BWA-MEM2/GATK.
- ▶ [Barbitoff et al](#) – Benchmark study comparing various alignment tools and variant callers
 - ▶ Variant callers have a much larger effect on variant than choice of aligner.
 - ▶ Mentioned that BWA-MEM2 is a reliable aligner
 - ▶ Bowtie2 shouldn't be used for human variant calling, significantly worse performance than other aligners.

Post-alignment processing

- ▶ Sorting and indexing bam files for downstream analysis
- ▶ Marking duplicates
 - ▶ Read duplicates give false confident in a specific variant and can affect copy number calling
- ▶ Base quality score recalibration
 - ▶ Sequencing techniques have certain biases for determining quality scores. Tools like GATK BaseRecalibrator help correct for this

Best practices during variant calling

- ▶ GATK (HaplotypeCaller) – Most popular tool for germline variant calling
 - ▶ Pros: Good sensitivity for SNVs and indels, BQSR included
 - ▶ Computationally very slow, requires multiple steps
- ▶ FreeBayes – Germline variant caller sed for small datasets
 - ▶ Pros: Can call complex variants, works well on non-model organisms
 - ▶ Slower than GATK, high false positive rate
- ▶ Strelka2 – Fast, mainly used for somatic variants
 - ▶ Pros: Fast and memory-efficient, high accuracy
 - ▶ Not well suited for structural variant detection

Best practices during variant calling

- ▶ Mutect2 (GATK) – Somatic variant calling in cancer
 - ▶ Pros: Good sensitivity for low-frequency SNVs and indels. Works well with matched normal-tumor samples
 - ▶ Computationally very slow and intensive, prone to false positives
- ▶ DeepVariant (Google) – Germline variant caller
 - ▶ Pros: Highly accurate due to deep learning approach, fewer false positives, open-source
 - ▶ Cons: Requires GPU acceleration for best performance
- ▶ DRAGEN (Illumina)
 - ▶ Pros: Very fast, integrated QC and filtering, alignment and variant calling all in one step
 - ▶ Cons: Requires license and expensive hardware dependency

Studies comparing variant callers

- ▶ [Betschart et al](#) - Benchmark study comparing GATK/BWA-MEM, DeepVariant, and DRAGEN for germline variant calling
 - ▶ GATK had the lowest performance of the three
 - ▶ However, DeepVariant requires GPU acceleration and DRAGEN requires a license
- ▶ [Barbitoff et al](#) – Benchmark study comparing several germline variant callers
 - ▶ Modern machine learning models like DeepVariant outperform traditional models like GATK, even when using GATK's machine learning model

Studies comparing variant callers

- ▶ [Abdelwehab et al](#) – Compared 5 germline variant callers for short and long reads
 - ▶ Short reads - DeepVariant provided the most balanced variant calling performance, GPU compatible. Highest F1 score for SNVs and INDELs
 - ▶ GATK4 was the most conventional tool, allows for manual filtering and thresholding
 - ▶ Long reads – DeepVariant outperformed conventional methods, especially for indels. DNAscope is a computationally efficient AI tool that also performed well.

Studies comparing variant callers

- ▶ [Guille et al](#) – Compared 20 somatic variant callers for WES datasets with reference mutations
 - ▶ Reads were aligned using BWA for all variant callers and base qualities were recalibrated using GATK
 - ▶ 20 tools were used including widely used tools, commercial tools, deep learning tools
 - ▶ Top performers for somatic SNVs: Dragen, Muse, TNScope, Mutect2
 - ▶ Top performers for somatic indels: NeuSomatic, DeepSomatic, DRAGEN
 - ▶ DRAGEN was the most robust and also was one of the fastest
 - ▶ Commercial tools generally outperformed open-source tools

Germline vs Somatic variants

- ▶ Different Assumptions: Germline variant callers (e.g., GATK HaplotypeCaller, DeepVariant) assume diploidy with allele frequencies of ~50% or 100%, while somatic callers (e.g., Mutect2, Strelka2) detect low-VOF mutations and require stricter filtering.
- ▶ Matched Normal Requirement: Somatic callers often need a matched normal sample to filter out inherited variants, whereas germline callers use population databases for filtering.
- ▶ Error Handling & Sensitivity: Somatic callers are optimized for low-frequency mutations and tumor heterogeneity, while germline callers focus on high-confidence inherited variants

Structural variants/Copy number variation

- ▶ [Gabrielaite et al.](#) – Study comparing 11 tools for copy-number variation detection in germline WGS and WES data.
- ▶ GATK gCNV had high recall but low precision.
- ▶ Lumpy and DELLY had more balanced recall and precision, reducing false positives.
- ▶ Manta and CNVkit performance varied a lot across datasets
- ▶ Study suggested that using multiple tools is the gold standard

Variant Filtering

- ▶ [Pedersen et al.](#) - Study evaluates variant filtering strategies using two WGS and WES family cohorts.
- ▶ Genotyping Quality (GQ ≥ 20) – ensures confidence in a genotype call
- ▶ Allele Balance (AB: 0.2-0.8) – Extreme values could suggest sequencing bias or error
- ▶ Sequencing Depth (DP ≥ 10) - Minimum sequencing depth covering a variant site to reduce false positives
- ▶ Population Allele Frequency (<0.001 for dominant and <0.01 for recessive)

Annotation

- ▶ **Functional Annotation:** Predicts the biological impact of variants using tools like SIFT, PolyPhen, and CADD to assess protein function and regulatory effects.
- ▶ **Population & Clinical Databases:** Filters variants based on population allele frequencies from gnomAD and 1000 Genomes and links to disease databases like ClinVar and COSMIC.
- ▶ **Pathogenicity:** Uses quality filters and pathogenicity scoring to identify rare, high-impact variants relevant to disease or functional studies.

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- ▶ GATK –genotype gVCF can genotype across samples.
 - ▶ Long read alignment, minimap